

Element-Based Floating Frame of Reference for Truss with Complex Geometry via Isoparametrize Spatial Beam

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1. Introduction

Floating Frame of Reference (FFRF) is a classical model for flexible multibody system dynamics. This method separates the rigid-body motion and flexible deformation under the assumption of small deformation. In this way, the flexible deformation can be easily reduced using vibration modes. This method is implemented in commercial software like ADAMS and RecurDyn, as it is easy to obtain modes' information from MNF files. However, it is still complicated if one wants to develop his own code for flexible bodies with complex geometry. This technique document aims to discuss one possible way to achieve cheap implementation by constructing shape integral of inertia only on element level.

2. Kinematic Notation

Popular FFRF methods use space $\mathbb{Q} = \mathbb{R}^3 \times \text{SO}(3)$ to model the rigid body motion. In this document, we follow the same representation. The displacement $\mathbf{r}$ of one point on the flexible body reads

$$\mathbf{r} = \mathbf{p} + \mathbf{R}(\bar{\mathbf{x}}_0 + \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_0)) \quad (1)$$

where $\mathbf{p}$ denotes the displacement of the reference point on the flexible body, $\mathbf{R}$ is rotation matrix of the reference point, $\bar{\mathbf{x}}_0$ is the static local position of points on the flexible body referred to the reference point, and $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_0)$ is the flexible deformation at point $\bar{\mathbf{x}}_0$ described in the reference frame $\mathbf{R}$. Notice that variables with $\bar{\cdot}$ are described in reference frame.

In this document, we use following augmented Hamilton's variational principle to formulate the dynamic equation,

$$\delta H = \delta \int_{t_0}^{t_1} (T - V + \Phi^T \lambda) dt = 0 \quad (2)$$

with T and V respectively the kinematic energy and elastic energy, Φ the constraint equation and λ the corresponding Lagrangian multiplier vector. Due to the small deformation assumption, the elastic energy can be calculated in the engineering way, which we do not further discuss. A more complicated term is the kinematic energy, as Eq. (1) shows the coupling of rigid body rotation and elastic deformation. To derive the kinematic energy, we take the time derivative of Eq. (1),

$$\dot{\mathbf{r}} = \dot{\mathbf{p}} + \dot{\mathbf{R}}(\bar{\mathbf{x}}_0 + \bar{\mathbf{u}}_f) + \mathbf{R}\dot{\bar{\mathbf{u}}}_f \quad (3)$$

The rotation matrix follows the left-invariant identity of SO(3) group, so that

$$\dot{\mathbf{R}} = \mathbf{R}\bar{\boldsymbol{\omega}}^\wedge \quad (4)$$

here $\bar{\boldsymbol{\omega}}^\wedge$ is an element of Lie algebra so(3), which is an asymmetric matrix, corresponding to angular velocity $\bar{\boldsymbol{\omega}}$. By using the identity of asymmetric matrix,

$$\dot{\mathbf{R}}(\bar{\mathbf{x}}_0 + \bar{\mathbf{u}}_f) = \mathbf{R}(\bar{\mathbf{x}}_0 + \bar{\mathbf{u}}_f)^\wedge \bar{\boldsymbol{\omega}} = \mathbf{R}\bar{\mathbf{d}}^\wedge \bar{\boldsymbol{\omega}} \quad (5)$$

Then Eq. (3) can be rewritten as

$$\dot{\mathbf{r}} = \begin{bmatrix} \mathbf{I} & \mathbf{R}\bar{\mathbf{d}}(\bar{\mathbf{x}}_0)^\wedge & \mathbf{R}\mathbf{S}(\bar{\mathbf{x}}_0) \end{bmatrix} \begin{bmatrix} \dot{\mathbf{p}} \\ \dot{\bar{\boldsymbol{\omega}}} \\ \dot{\bar{\mathbf{e}}} \end{bmatrix} = \mathbf{L}\dot{\mathbf{q}} \quad (6)$$

where $\bar{\mathbf{e}}$ stands for the generalized coordinates of deformation, and $\mathbf{S}(\bar{\mathbf{x}}_0)$ is the shape functions of deformation, or, it can also be regarded as mode functions. The corresponding Kinematic Energy then reads

$$T = \frac{1}{2} \dot{\mathbf{q}}^T \int_V \mathbf{L}^T \mathbf{L} dV \dot{\mathbf{q}} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} \quad (7)$$

Notice that $\mathbf{L}$ contains not only local static position $\bar{\mathbf{x}}_0$ but also its corresponding shape function $\mathbf{S}(\bar{\mathbf{x}}_0)$ together the deformation coordinates $\bar{\mathbf{e}}$. We unfold the middle term $\mathbf{L}^T \mathbf{L}$

$$\mathbf{L}^T \mathbf{L} = \begin{bmatrix} \mathbf{I} & \mathbf{Rd}(\bar{\mathbf{x}}_0)^\wedge & \mathbf{RS}(\bar{\mathbf{x}}_0) \\ & -\mathbf{d}(\bar{\mathbf{x}}_0)^\wedge \mathbf{d}(\bar{\mathbf{x}}_0)^\wedge & -\mathbf{d}(\bar{\mathbf{x}}_0)^\wedge \mathbf{S}(\bar{\mathbf{x}}_0) \\ sym & & \mathbf{S}(\bar{\mathbf{x}}_0)^T \mathbf{S}(\bar{\mathbf{x}}_0) \end{bmatrix} \quad (8)$$

For one flexible body with complex geometry like some truss structure, it is hard to directly conduct such shape integral by using Eq. (8). Moreover, most engineering finite element methods use locally isoparametrized shape function instead of globalized shape function, as the global mode functions are often not known.

3. Isoparameterized Spatial Beam

To avoid this problem, we propose an element-based method to avoid global shape integral. Without the loss of generality, we take this isoparameterized spatial beam for example, as illustrated in Fig. 1. The axial direction of one single beam element is denoted as $\bar{\mathbf{e}}_\xi^b$, with two orthogonal unit vectors $\bar{\mathbf{e}}_\zeta^b$ and $\bar{\mathbf{e}}_\tau^b$ describe the cross-section. Then, the static position on the axis of one beam element can be described using

$$\bar{\mathbf{x}}_c(\xi) = \frac{L-\xi}{L} \bar{\mathbf{x}}_i + \frac{\xi}{L} \bar{\mathbf{x}}_j \quad (9)$$

Then arbitrary points on the beam element can be obtained with

$$\bar{\mathbf{x}}_0(\xi, \zeta, \tau) = \bar{\mathbf{x}}_c(\xi) + \mathbf{O}_0 \bar{\mathbf{y}}(\zeta, \tau) \quad (10)$$

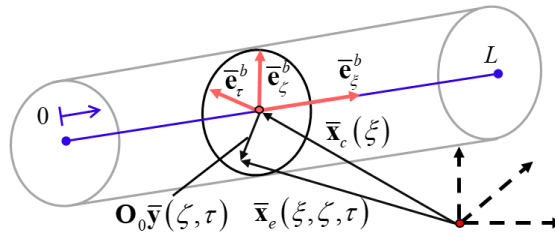


Fig. 1. Isoparameterized Spatial Beam

To describe the exact deformation of small deformation beam, the deformation on arbitrary point of beam element reads

$$\bar{\mathbf{u}}_f = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} u_c - \zeta \frac{\partial v_c}{\partial x} - \tau \frac{\partial w_c}{\partial x} \\ v_c - \tau \varphi \\ w_c + \zeta \varphi \end{bmatrix} \quad (11)$$

where u , v and w stand for deformation in three directions, u_c , v_c and w_c stands for the axial stretch and deflection in two directions, φ is the torsion angle.

According to Euler-Bernoulli beam theory, the shape function can be written as

$$\mathbf{S}(\xi, \zeta, \tau) = \begin{bmatrix} S_{11}(\xi) & -\zeta S'_{21}(\xi) & -\tau S'_{21}(\xi) & 0 & \tau S'_{31}(\xi) & -\zeta S'_{31}(\xi) & S_{12}(\xi) & -\zeta S'_{22}(\xi) & -\tau S'_{22}(\xi) & 0 & \tau S'_{32}(\xi) & -\zeta S'_{32}(\xi) \\ 0 & S_{21}(\xi) & 0 & -\tau S_{11}(\xi) & 0 & S_{31}(\xi) & 0 & S_{22}(\xi) & 0 & -\tau S_{12}(\xi) & 0 & S_{32}(\xi) \\ 0 & 0 & S_{21}(\xi) & \zeta S_{11}(\xi) & -S_{31}(\xi) & 0 & 0 & 0 & S_{22}(\xi) & \zeta S_{12}(\xi) & -S_{32}(\xi) & 0 \end{bmatrix} \quad (12)$$

The detailed shape function $S_{ij}(\xi)$ follows the classical shape function of Euler-Bernoulli beam. Then the shape integral over whole flexible body can be separated into element-level integral, i.e.

$$\mathbf{M}_i = \int_{V^i} \mathbf{L}(\xi, \zeta, \tau)^T \mathbf{L}(\xi, \zeta, \tau) dV^i = \begin{bmatrix} \mathbf{M}_{tt,i} & \mathbf{M}_{tr,i} & \mathbf{M}_{tf,i} \\ & \mathbf{M}_{rr,i} & \mathbf{M}_{rf,i} \\ sym & & \mathbf{M}_{ff,i} \end{bmatrix} \quad (13)$$

Notice that the all elements share the same rigid body displacement and rigid body rotation, so that when we use bool operation to assembly $\mathbf{M}_i$, the corresponding rigid body terms can be sum together. Such bool operation can be defined as

$$\mathbf{M} = \text{bool}_{i=1}^n(\mathbf{M}_i) = \begin{bmatrix} \sum_{i=1}^n \mathbf{M}_{tt,i} & \sum_{i=1}^n \mathbf{M}_{tr,i} & \mathbf{M}_{tf,i} \mathbf{T}_i \\ & \sum_{i=1}^n \mathbf{M}_{rr,i} & \mathbf{M}_{rf,i} \mathbf{T}_i \\ sym & & \mathbf{T}_i^T \mathbf{M}_{rr,i} \mathbf{T}_i \end{bmatrix} \quad (14)$$

where $\mathbf{T}_i$ is the assembly operator in classical FEM. In this way, the shape integral is separated on single element. From the perspective of code developing, complex geometry is no longer a problem.